

Pre-trained Foundation Models as Fast Proxies for Flow and Transport in Porous Media

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Topic

Fast proxy models can reduce the cost of repeated subsurface simulations for uncertainty quantification, optimization, and data assimilation. Conventional neural surrogates are typically trained for a specific problem and require substantial high-fidelity simulation data. PDE foundation models instead learn from diverse physical systems before being fine-tuned on a smaller dataset. Examples include Poseidon [1], multiple physics pretraining (MPP) [2], DPOT [3], and Walrus [4].

Porous media problems combine pressure equations with transport dynamics, sharp fronts, heterogeneous rock properties, and wells. Transferring learned representations to this setting may be challenging, including the risk of negative transfer between flow regimes [5]. This project asks: *does pre-training improve accuracy and data efficiency compared with training a dedicated neural operator from scratch?* Using open-DARTS [6] to generate reference data, the study will compare fine-tuned foundation models with FNO and U-FNO [7] baselines.

Tasks

- 1. Generate data:** Simulate tracer transport and two-phase displacement on structured 2D grids with Gaussian and channelized permeability fields; vary rock properties and well controls.
- 2. Train and adapt:** Fine-tune pre-trained Poseidon checkpoints for the 2D cases, using rock properties and well controls as inputs. Train FNO/U-FNO baselines from scratch on the same data. Use pre-trained PyTorch checkpoints on department servers or DelftBlue GPUs.
- 3. Evaluate:** Measure accuracy versus training-set size, front position, mass balance error, long-term rollout stability, generalization to unseen geology and well controls, and inference speed-up relative to open-DARTS.
- 4. Optional extensions:** Explore 3D CO₂ injection or geothermal cases with suitable DPOT/Walrus checkpoints, or apply the best proxy to data assimilation.

Requirements

Good Python programming skills and familiarity with flow and transport in porous media; PyTorch experience is an advantage.

References

- [1] M. Herde et al. “Poseidon: Efficient Foundation Models for PDEs”. In: *Advances in Neural Information Processing Systems*. 2024.
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- [4] M. McCabe et al. *Walrus: A Cross-Domain Foundation Model for Continuum Dynamics*. arXiv:2511.15684. 2025.
- [5] E. Sharma et al. *Eradicating Negative Transfer in Multi-Physics Foundation Models via Sparse Mixture-of-Experts Routing*. arXiv:2605.15179. 2026.
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